

HR Analytics & Workforce Attrition Analysis

Comprehensive Diagnostic Report & Predictive Machine Learning Insights

Scope: Enterprise Workforce Analysis

Total Records: 1,470 Employees

Target Metric: Voluntary & Involuntary Attrition

Status: Executive Summary

TOTAL STAFF
1,470
Active Base

ATTRITION COUNT
237
Total Departures

ATTRITION RATE
16.12%
Baseline Metric

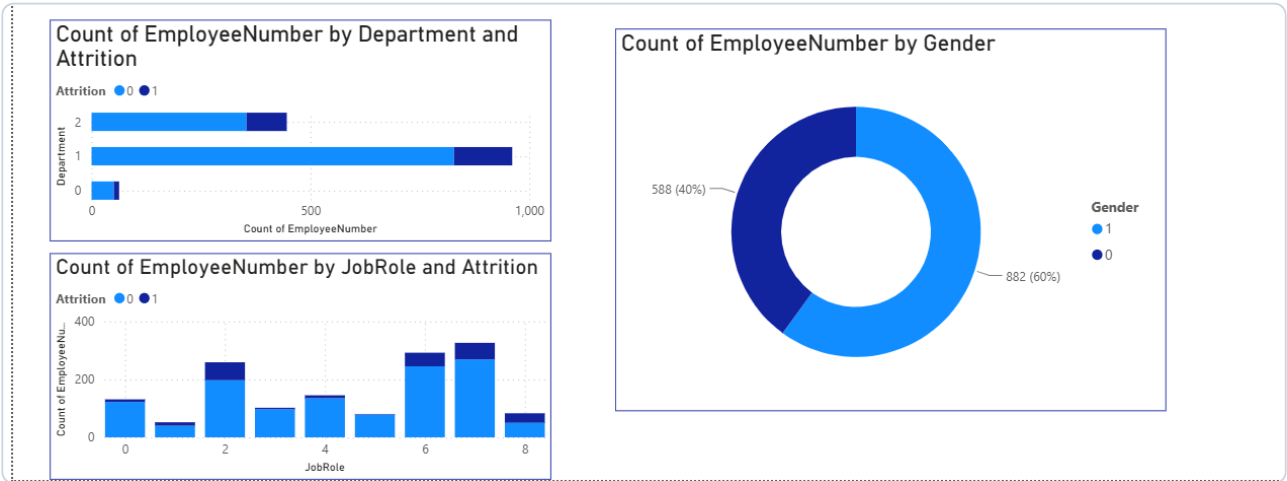
AVERAGE AGE
36.9
Years Old

AVG INCOME
\$6,503
Per Month

AVG TENURE
7.0
Years at Firm

1. EXECUTIVE SUMMARY & ORGANIZATIONAL DEMOGRAPHICS

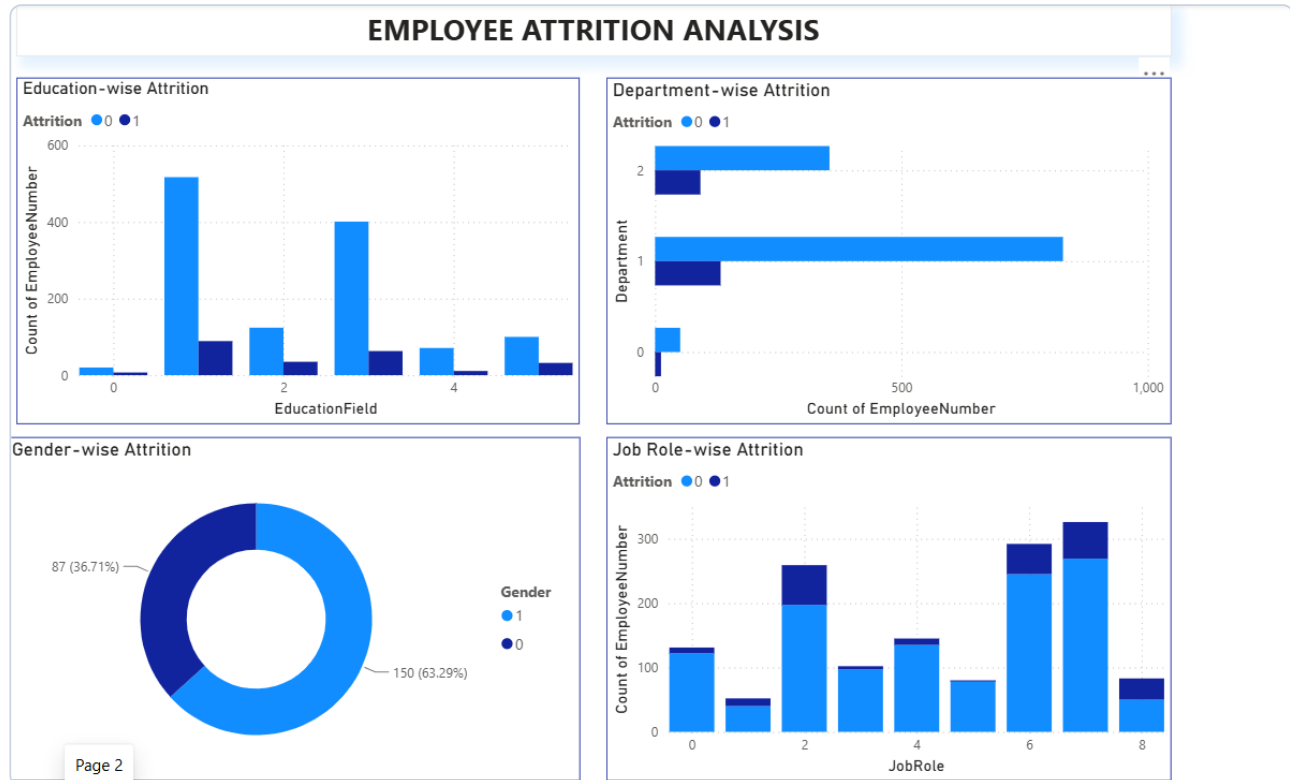
This analysis provides a data-driven evaluation of employee attrition across key departments, roles, demographic profiles, and compensation tiers. With an overall turnover rate of **16.12%** (237 departures out of 1,470 total staff), target interventions are necessary to retain core technical and sales talent.



Demographic Variable	Observations & Findings	Risk Impact
Departmental Breakdown	Research & Development exhibits the largest volume of employees, while Sales experiences higher proportional attrition relative to its size.	High
Gender Ratio	Workforce is comprised of approximately 60% male (882) and 40% female (588) employees. Turnover distribution closely mirrors this baseline ratio.	Moderate
Job Role Concentration	Sales Representatives, Laboratory Technicians, and Research Scientists account for the vast majority of turnover cases.	Critical

2. DEPARTMENTAL & ROLE-WISE ATTRITION BREAKDOWN

Detailed inspection across field backgrounds and job titles reveals concentrated turnover pools. Educational specialization in Life Sciences and Medical fields represents the largest baseline, but technical roles show elevated flight risk.

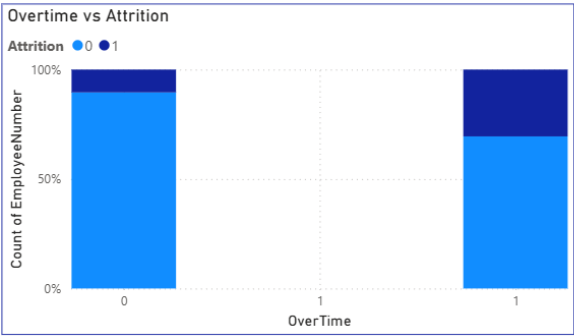
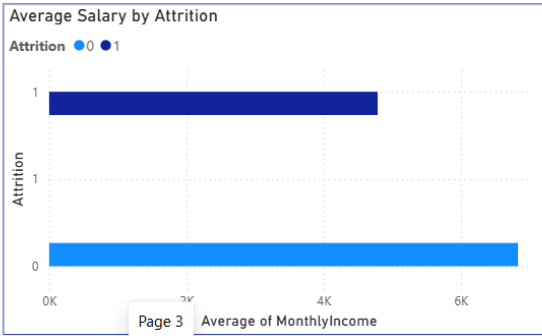
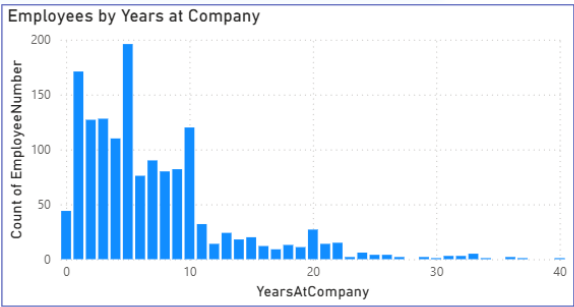
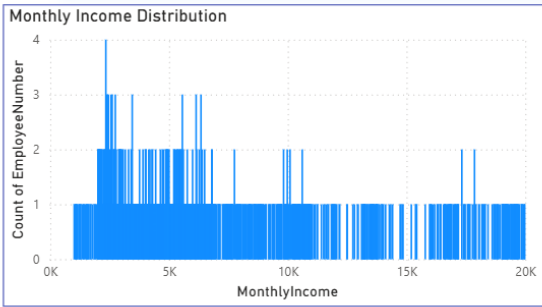


- **Education Field Impact:** Employees with backgrounds in Life Sciences and Medical fields comprise over 70% of total headcount. High absolute turnover occurs in these categories, but Marketing and Technical Degrees exhibit higher percentage attrition rates.
- **Critical Role Hotspots:** Entry-level roles such as Sales Representatives and Lab Technicians show distinct attrition spikes, pointing to onboarding friction and early-career compensation gaps.

3. COMPENSATION, TENURE & OVERTIME DYNAMICS

Financial compensation and workload pressures represent two of the strongest direct drivers of employee departures.

SALARY & EXPERIENCE ANALYSIS



Salary Disparity Impact:

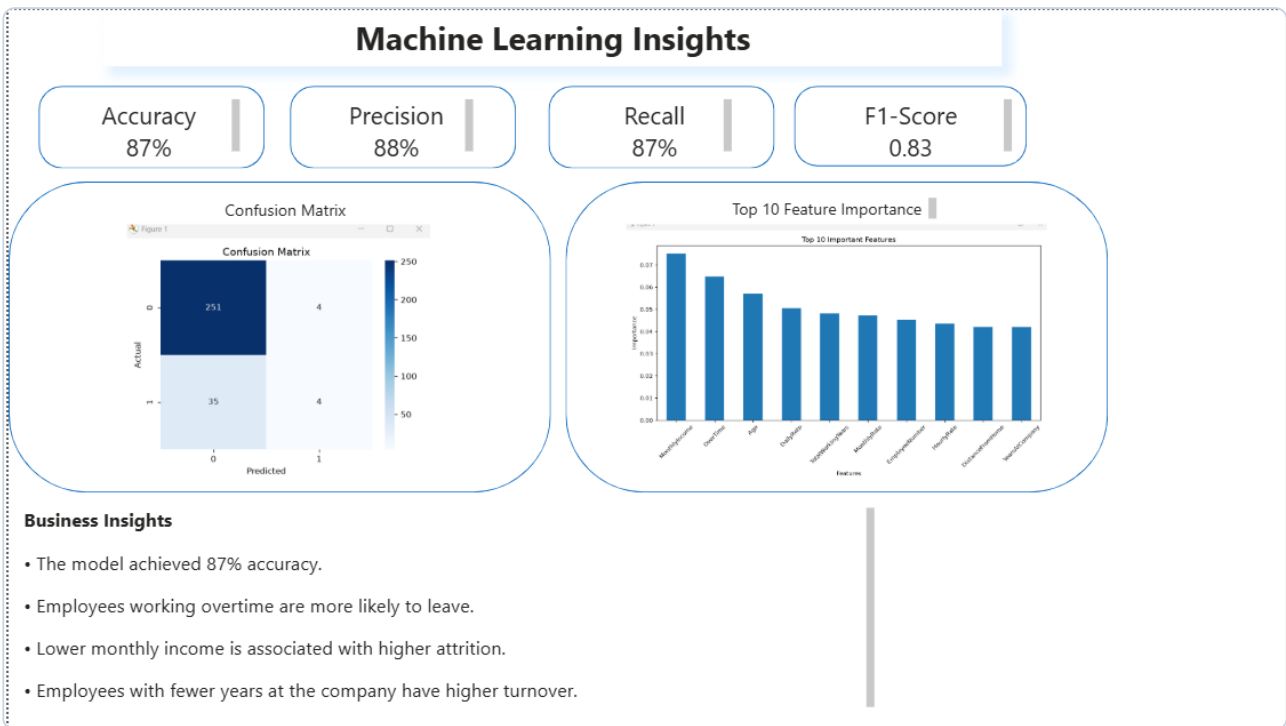
Departing staff average significantly lower monthly earnings (~\$4,800) compared to retained staff (~\$6,800). Lower salary brackets face severe flight risk.

Overtime Burnout Factor:

Employees working regular Overtime exhibit an attrition rate exceeding 30%, whereas non-overtime staff maintain attrition below 10%.

4. PREDICTIVE MACHINE LEARNING INSIGHTS

A machine learning classification model was trained to predict employee flight risk. The model achieved high overall predictive performance across core validation metrics:



Model Metric	Value	Strategic Interpretation
Accuracy	87%	Overall correctness across both retention and turnover predictions.
Precision	88%	High precision minimizes false positives when flagging flight-risk employees.
Recall	87%	Effectively captures the vast majority of actual employee departures.
F1-Score	0.83	Strong harmonic balance between precision and recall for actionable HR alerting.

Primary Drivers of Attrition (Feature Importance):

1. **Monthly Income:** Primary financial determinant influencing stay-vs-leave decisions.
2. **Overtime:** Direct operational catalyst driving burnout and sudden resignations.
3. **Age & Total Working Years:** Younger, early-career staff are more mobile and prone to changing companies.
4. **Distance From Home:** Long daily commutes significantly accelerate attrition over time.
5. **Years At Company:** Critical flight risk peak occurs within Years 1–2 of service.

5. STRATEGIC RECOMMENDATIONS & ACTION PLAN

- **Workload & Overtime Management:** Implement strict caps on sustained overtime for high-risk job roles (Sales Reps and Lab Technicians). Re-evaluate team headcount allocation in R&D.
- **Targeted Compensation Adjustments:** Conduct benchmark compensation audits for Job Levels 1 and 2 to eliminate pay disparities that drive early-career departures.

- **Structured 30-60-90 Onboarding:** Strengthen early-tenure engagement and career path clarity to mitigate high turnover during the first 24 months of tenure.
- **Flexible & Hybrid Work Options:** Introduce remote or flexible scheduling policies for employees facing long commutes (>15 miles from home).